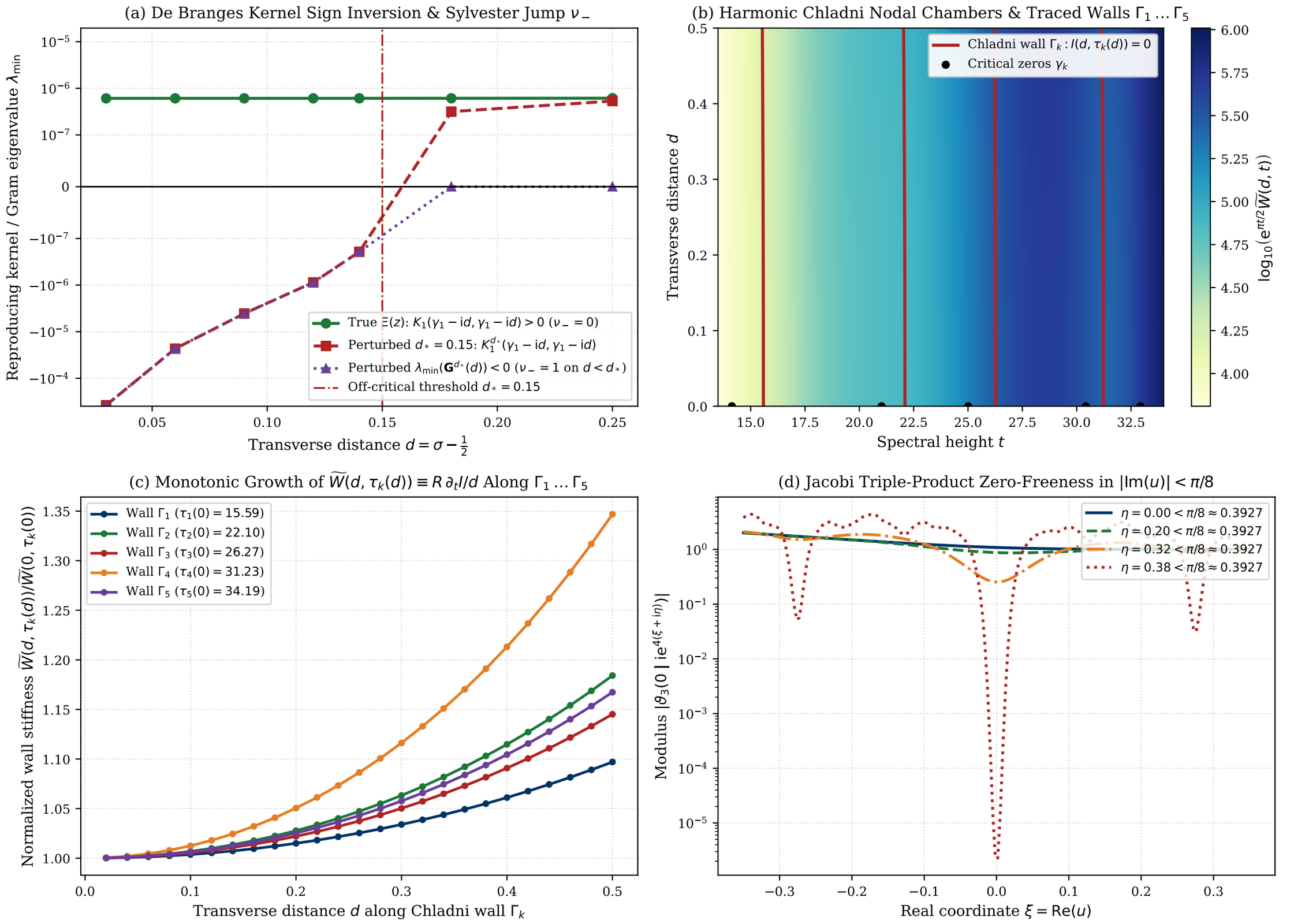

(a) De Branges Kernel Sign Inversion & Sylvester Jump ν_-
Reproducing kernel / Gram eigenvalue $\lambda_{\min}$
Transverse distance $d = \sigma - \frac{1}{2}$
True $\Xi(z)$: $K_1(\gamma_1 - id, \gamma_1 - id) > 0$ ($\nu_- = 0$)
Perturbed $d_* = 0.15$: $K_1^{d_*}(\gamma_1 - id, \gamma_1 - id)$
Perturbed $\lambda_{\min}(\mathbf{G}^{d_*}(d)) < 0$ ($\nu_- = 1$ on $d < d_*$)
Off-critical threshold $d_* = 0.15$
(b) Harmonic Chladni Nodal Chambers & Traced Walls $\Gamma_1 \dots \Gamma_5$
Transverse distance d
Spectral height t
Chladni wall $\Gamma_k : I(d, \tau_k(d)) = 0$
Critical zeros γ_k
$\log_{10}(e^{\pi t/2}\widetilde{W}(d, t))$
(c) Monotonic Growth of $\widetilde{W}(d, \tau_k(d)) \equiv R\,\partial_t I/d$ Along $\Gamma_1 \dots \Gamma_5$
Normalized wall stiffness $\widetilde{W}(d, \tau_k(d))/\widetilde{W}(0, \tau_k(0))$
Transverse distance d along Chladni wall Γ_k
Wall Γ_1 ($\tau_1(0) = 15.59$)
Wall Γ_2 ($\tau_2(0) = 22.10$)
Wall Γ_3 ($\tau_3(0) = 26.27$)
Wall Γ_4 ($\tau_4(0) = 31.23$)
Wall Γ_5 ($\tau_5(0) = 34.19$)
(d) Jacobi Triple-Product Zero-Freeness in $|\mathrm{Im}(u)| < \pi/8$
Modulus $|\vartheta_3(0 \mid ie^{4(\xi + i\eta)})|$
Real coordinate $\xi = \mathrm{Re}(u)$
$\eta = 0.00 < \pi/8 \approx 0.3927$
$\eta = 0.20 < \pi/8 \approx 0.3927$
$\eta = 0.32 < \pi/8 \approx 0.3927$
$\eta = 0.38 < \pi/8 \approx 0.3927$